

Sericulture Management App Specification

Overview

A mobile and web-based sericulture management app for:

- Eri silkworm rearing
- Bombyx mori silkworm rearing
- Predictive lifecycle tracking
- Feeding schedule management
- Environmental monitoring
- Cocoon production analytics
- Batch performance reporting

The app should support both small-scale and commercial sericulture operations.

Core Objectives

The app should help a farmer or rearing technician: 1. Plan and track rearing cycles 2. Predict important lifecycle dates automatically 3. Record feeding schedules and quantities 4. Monitor larval growth stages 5. Record mortality and disease incidents 6. Analyze cocoon quality and yield 7. Compare performance across batches 8. Generate reports for productivity and profitability

Species Support

1. Eri Silkworm

Support for:

- Samia ricini lifecycle
- Castor leaf feeding
- Kesseru feeding (optional)
- Eri-specific moulting schedules
- Open rearing practices

2. Bombyx Mori

Support for:

- Mulberry feeding schedules
- Instar-based rearing

- Temperature and humidity tracking
 - Mounting schedules
 - Cocoon harvest timing
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Main Modules

1. Dashboard

The dashboard should display:

- Active rearing batches
- Current instar stage
- Next feeding time
- Upcoming moult dates
- Expected spinning dates
- Expected cocoon harvest dates
- Mortality alerts
- Environmental alerts
- Batch performance summaries

Dashboard Widgets

- Total worms
 - Survival rate
 - Average cocoon weight
 - Feed consumption
 - Rearing duration
 - Expected yield
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2. Batch Management

Each rearing operation should be treated as a batch.

Batch Details

Fields:

- Batch ID
- Species (Eri or Bombyx)
- Strain
- Start date
- Egg source

- Number of eggs/larvae
- Rearing house/location
- Caretaker name
- Feeding material

Batch Status

- Active
 - Mounting
 - Cocooning
 - Harvested
 - Closed
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3. Predictive Lifecycle Engine

The app should automatically calculate and predict:

- Hatch date
- Each instar start/end date
- Molt dates
- Mounting date
- Spinning date
- Cocoon harvest date
- Moth emergence date

Prediction Inputs

- Species
- Temperature
- Humidity
- Strain
- Feeding frequency
- Historical performance

Example Prediction Logic

For Bombyx:

- Egg incubation: 10–12 days
- 1st instar: 3–4 days
- 2nd instar: 3 days
- 3rd instar: 4 days
- 4th instar: 5 days
- 5th instar: 8 days
- Spinning: 3 days

- Cocoon maturation: 5–7 days

The system should adjust predictions dynamically based on real observations.

4. Feeding Schedule Module

Features

- Feeding reminders
- Feed quantity logging
- Feed quality notes
- Missed feeding alerts
- Leaf inventory tracking

Feeding Records

Fields:

- Date/time
- Batch
- Feed type
- Feed quantity
- Feeding stage
- Notes

Smart Features

- Auto-suggest feeding intervals
 - Predict required leaf quantity
 - Compare feed efficiency
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5. Environmental Monitoring

Parameters

- Temperature
- Humidity
- Ventilation
- Light conditions

Features

- Manual entry
- Bluetooth sensor integration (future)
- Alerts when conditions exceed thresholds

Example Alerts

- High humidity warning
 - Temperature stress risk
 - Disease risk indicator
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6. Health and Mortality Tracking

Records

- Daily mortality count
- Disease symptoms
- Treatment applied
- Isolation records

Disease Library

Support for:

- Flacherie
 - Grasserie
 - Muscardine
 - Pebrine
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7. Cocoon Analysis Module

This module will analyze cocoon quality and production.

Data Inputs

- Number of cocoons
- Total cocoon weight
- Defective cocoons
- Shell weight
- Filament length
- Renditta

- Moisture content
- Cocoon dimensions

Calculated Metrics

- Average cocoon weight
- Shell ratio
- Survival percentage
- Yield per 100 disease-free layings
- Defect rate
- Reelability index
- Production efficiency

Formula Engine

The app should support custom formulas provided by the user.

Formula Capabilities

- User-defined equations
- Multiple variables
- Auto-calculation
- Historical comparisons
- Exportable analysis reports

Example: Average Cocoon Weight = Total Cocoon Weight / Number of Cocoons

8. Reports and Analytics

Reports

- Batch performance
- Feeding efficiency
- Mortality trends
- Cocoon quality reports
- Environmental trends
- Production forecasting

Charts

- Growth curves
- Feed consumption charts
- Mortality charts
- Yield trends
- Temperature vs performance

Export Options

- PDF
 - Excel
 - CSV
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9. Notifications and Alerts

Notification Types

- Feeding reminders
 - Moulting alerts
 - Harvest reminders
 - Environmental warnings
 - Disease risk alerts
 - Task reminders
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10. Multi-User Support

Roles:

- Farm owner
- Supervisor
- Technician
- Data entry operator

Permissions:

- View only
 - Edit batches
 - Manage analytics
 - Export reports
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Technical Architecture

Recommended Stack

Mobile App

- Flutter (Android + iOS)

Backend

- Node.js or Django

Database

- PostgreSQL

Cloud

- Firebase or AWS

Analytics

- Python analytics engine
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Offline Capability

The app should support:

- Offline data entry
- Sync when internet becomes available
- Local notifications

This is important for rural farming areas.

AI/Smart Features (Advanced)

Future enhancements:

- Disease detection from images
 - Cocoon grading from photos
 - Predictive mortality risk
 - Feed optimization suggestions
 - Production forecasting
 - AI chatbot assistant for farmers
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Suggested Screens

1. Login
2. Dashboard
3. Create Batch

4. Batch Details
 5. Feeding Logs
 6. Lifecycle Calendar
 7. Environment Monitor
 8. Cocoon Analysis
 9. Reports
 10. Settings
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Database Structure (Simplified)

Tables

Users

- id
- name
- role
- phone

Batches

- id
- species
- strain
- start_date
- egg_count
- status

FeedLogs

- id
- batch_id
- datetime
- feed_type
- quantity

EnvironmentLogs

- id
- batch_id
- temperature
- humidity
- timestamp

MortalityLogs

- id
- batch_id
- count
- reason

CocoonAnalysis

- id
 - batch_id
 - cocoon_count
 - total_weight
 - shell_weight
 - custom_metrics
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Suggested MVP (Version 1)

Essential Features

- Batch management
 - Lifecycle prediction
 - Feeding schedule logging
 - Notifications
 - Cocoon analysis
 - Basic reporting
 - Offline support
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Future Versions

Version 2

- Sensor integration
- Photo analysis
- AI disease detection
- Multi-farm management

Version 3

- Marketplace integration
- Buyer management
- Financial tracking
- Supply chain management

What You Can Provide Next

To continue designing the app, you can provide:

1. Your cocoon analysis formulas
2. Preferred platform (Android only or Android + iPhone)
3. Whether you want offline-first support
4. Sample rearing schedules
5. Sample farm records
6. Any local language support requirements
7. Whether this is for personal use or commercial deployment

Once you provide the formulas, the cocoon analysis engine can be designed precisely around your workflow.